Zihang Huang

Autonomous Driving Perception Leader | Principal Engineer & Discipline Manager | D.Eng. Candidate

Email: derekwong166@gmail.com | **Phone:** (+65)84265804 | **Singapore PR** | **HK QMAS Holder**

Professional Summary

R&D Leader and D.Eng. Candidate with 7+ years of experience leading autonomous driving perception initiatives across L2+ mass production and L4 platform development. Managed **cross-regional teams of 40+ engineers** across Singapore, Shenzhen, Chengdu, Nanjing, and Huizhou, defining technical roadmaps, KPI frameworks, and delivery plans for perception systems. Delivered **Dongfeng eπ007/eπ008 L2+ start of production (SOP)**, initiated an L4 unmanned logistics vehicle perception platform, and architected **data-loop infrastructure that reduced annotation cost by 99%**. Focused on building high-performance technical organizations while bridging BEV perception, multi-modal sensor fusion, embedded AI deployment, and VLM/VLA-based multimodal AI systems.

Experience

Nov. 2025 - Feb. 2026 — **Anker Technology Singapore Pte. Ltd.**

Senior Algorithm Engineer

Project: Multimodal Customer Feedback Diagnosis Agent for robot vacuum products

- Architected and deployed an end-to-end **LLM/VLM-based multimodal diagnostic pipeline** for robot vacuum failure analysis, integrating ROS bags, telemetry logs, scene images, and customer feedback.
- Developed a **multi-agent reasoning** framework for joint analysis of unstructured customer feedback and structured robot runtime data, achieving **80% automated diagnosis accuracy** and reduced per-case analysis time by **over 85%**.

Apr. 2022 - Oct. 2025 — **Desay SV Automotive Singapore Pte. Ltd.**

Principal Software Engineer / Discipline Manager

- **Cross-Regional Engineering Leadership:** Orchestrated a distributed perception R&D organization of **40+ engineers** across Singapore, Shenzhen, Chengdu, Nanjing, and Huizhou. Defined technical roadmaps, KPI frameworks, and cross-site delivery mechanisms across L2+ mass production, L4 platform development, and data-loop infrastructure.
- **L2+ Mass Production Delivery:** Directed end-to-end perception stack development and deployment for Dongfeng eπ007/eπ008, managing technical risks and resource allocation to ensure successful **start of production (SOP) in June 2024**.
- **Data-Loop & MLOps Innovation:** Drove R&D operational efficiency by designing an industrial-grade **auto-labeling and scenario reconstruction system** for perception data generation, validation, and model iteration. Reduced annotation cost by **99%** while scaling throughput to **192K frames/day** with **98.5% recall**.
- **L4 Autonomous Platform:** Championed the technical strategy for a new **L4 unmanned logistics vehicle** business line, architecting the perception framework from sensor suite configuration and multi-modal fusion to calibration, perception integration, and **prototype delivery in March 2025**.
- **Talent & Technical Bar-Raising:** Served as l**ead technical interviewer and mentor** for senior engineers, improving hiring quality and establishing architectural review standards across perception teams.

Apr. 2020 - Mar. 2022 — **Desay SV Automotive Singapore Pte. Ltd.**

Senior Software Engineer / Group Leader

- Directed a **7-engineer team** through **full-stack autonomous driving perception system** integration and validation, contributing to a Silver Medal result among 143 global teams at the 2021 World Intelligent Driving Challenge.
- Spearheaded algorithmic optimization for a **lightweight monocular 3D object detection engine** on the NVIDIA Xavier platform, achieving real-time inference with less than 5% position error under tight compute constraints and enabling deployment on resource-limited hardware.

Aug. 2018 - Mar. 2020 — **Desay SV Automotive Singapore Pte. Ltd.**

Software Engineer, Computer Vision (ISO 26262 Certified)

- Developed and deployed embedded computer vision models, including one-stage object detection with 95% mAP and monocular depth estimation with less than 3.4% error.
- Supported perception system development and validation for **Singapore LTA autonomous driving road-test approval,** one of the first international road-test licenses awarded to a Chinese Tier-1 supplier.

Education

2023 - Present — **South China University of Technology, School of Computer Science and Engineering, Guangzhou, China**
Doctor of Engineering (D.Eng.) Candidate, Focus on Industrial Autonomous Systems, Supervised by Professor C.L. Philip Chen, IEEE Fellow, Dean of the School of Computer Science & Engineering.

2016 - 2018 — **University of Pittsburgh, Swanson School of Engineering, Pittsburgh, PA, U.S.**
Master of Electrical and Computer Engineering **GPA:3.9/4.0**

2012 - 2016 — **Guangzhou University, School of Mechanical and Electrical Engineering, Guangzhou, China**
Bachelor of Electrical Engineering

Patents

Z. Huang, et al. *Category Determination Method, Apparatus, Device and Storage Medium*, Chinese Invention Patent, Application No. **CN115774844A**, Published: **Mar. 10, 2023**.

Z. Huang, et al. *3D Object Detection Method Based on Monocular Camera*, PCT Application**: PCT/CN2021/102534**, Filed: **Jun. 25, 2021**. Chinese Patent Application No. **CN112883790A**, Published: **Jun. 1, 2021**

M. Y. Yu, **Z. Huang**, et al. *Traffic Signal Light Detection Method, Apparatus, Device and Storage Medium for Vehicles*, Chinese Invention Patent, Application No. **CN117456509A**, Published: **Jan. 26, 2024**.

N. N. Liu, **Z. Huang**, et al. *Lane Line Tracking Method, Apparatus, Device and Medium*, Chinese Invention Patent, Application No. **CN117475394A**, Published: **Jan. 30, 2024**.

Y. X. Li, **Z. Huang**, et al. *Perception Fusion System, Electronic Device and Storage Medium*, Chinese Invention Patent, Application No. **CN116664997A**, Published: **Aug. 29, 2023**.

N. N. Liu, **Z. Huang**, et al. *Object Detection Method, Apparatus, Device and Storage Medium*, Chinese Invention Patent, Application No. **CN115761698A**, Published: **Mar. 7, 2023**.

X. H. Ge, **Z. Huang**, et al. *Vehicle Driving Control Method, System and Storage Medium*, Chinese Invention Patent, Application No. **CN116215526A**, Published: **Jun. 6, 2023**.

Siegfried Bolong, H. N. Ruan, **Z. Huang**, et al. *Automatic Calibration Method, System and Computer-Readable Storage Medium*, Chinese Invention Patent, Application No. **CN118376986A**, Published: **Jul. 23, 2024**.

Publications

Li, Yuxin, Yiheng Li, Xulei Yang, Mengying Yu, **Zihang Huang**, Xiaojun Wu, and Chai Kiat Yeo. *QuadBEV: An Efficient Quadruple-Task Perception Framework via Bird's-Eye-View Representation*. **2024 IEEE 27th International Conference on Intelligent Transportation Systems (ITSC)**, September 24- 27, 2024, Edmonton, Canada. DOI: https://doi.org/10.48550/arXiv.2410.06516

Li, Yuxin, Qiang Han, Mengying Yu, Yuxin Jiang, Chai Kiat Yeo, Yiheng Li, **Zihang Huang**, Nini Liu, Hsuanhan Chen, and Xiaojun Wu. *Towards Efficient 3D Object Detection in Bird's-Eye-Space for Autonomous Driving: A Convolutional-Only Approach*. **2023 IEEE 26th International Conference on Intelligent Transportation Systems (ITSC)**, pp. 2170-2177. DOI: https://doi.org/10.48550/arXiv.2312.00633

Li, Yuxin, Yiheng Li, Xulei Yang, Mengying Yu, **Zihang Huang**, Xiaojun Wu, and Chai Kiat Yeo. *Learning Content-Aware Multi-Modal Joint Input Pruning via Bird's-Eye-View Representation*. arXiv preprint arXiv:2410.07268 (2024). DOI: https://doi.org/10.48550/arXiv.2410.07268

K.-J. Wang, H.-W. Tung, **Z. Huang,** P. Thakur, Z.-H. Mao, M.-X. You. 2018. *EXGbuds: Universal Wearable Assistive Device for Disabled People to Interact with the Environment Seamlessly*, In **HRI'18 Companion**, March 5-8, 2018, Chicago, IL, USA. ACM, NY, IEEE, NY, USA, 2018. pp. 369–370. DOI: https://doi.org/10.1145/3173386.3177836

Awards and Honors

2022-2024	Desay SV Superstar Award - Excellent Project Team. Led perception part of the Smart Solution, ICP Aurora, and A1000 Algorithm teams to win consecutive annual company-wide top honors for delivery excellence.
2021	Silver Medal, World Intelligent Driving Challenge (2021): Ranked 6th/143 global teams in the competition.
2019	Innovation Star - First Prize: Recognized for breakthrough contributions in L3 Autonomous Driving key technologies.

Skills and Qualifications

Programming Languages: C++ (Modern C++11/14/17), Python, Shell Scripting.

Deep Learning & Frameworks: PyTorch, TensorFlow, TensorRT, CUDA, ONNX, OpenCV.

Robotics & Middleware: ROS / ROS2, Docker, Git, CMake, GDB, Linux System Programming.

Hardware Platforms: NVIDIA Orin / Xavier / Thor /BST series / Qualcomm

Algorithms: Multi-modal Sensor Fusion (Lidar/Camera/Radar), BEV Perception, 3D Object Detection, Kalman Filtering, Occupancy Networks, Vision-Language-Action, Vision-Language-Model.

Languages: English (Professional Proficiency), Mandarin (Native).

Research Interests

Autonomous Driving, Computer Vision, Deep Learning, Object Detection, Depth Estimation, Sensor Fusion.